

Jiacan Li

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🌐 Personal Page | 📁 GitHub Repository

Research Interests: Dexterous Manipulation, Robot Learning, 3D Perception, and Autonomous Navigation

Education

Southern University of Science and Technology

B.Eng. in Robotics Engineering

Department of Mechanical and Energy Engineering

• **GPA: 3.79/4.0**; weighted average: **91.37/100**.

• **Core Courses (scores out of 100):** Mechatronic Systems (98), Robot Operating System (99), Robot Modeling and Control (95), Computer Vision (94), Machine Learning for Engineering (97), Robotic Actuation System (94), Finite Element Theory and Its Engineering Applications (97), Fundamentals of Sensing Technology (92), Engineering Mechanics - Statics and Dynamics (93), Mechanical Design (97), CAD & Engineering Drawing (94).

Sep 2023 – Present

Shenzhen, China

National University of Singapore

Visiting Research Student

Department of Electrical and Computer Engineering

Jun 2026 – Present

Singapore

Publications & Patents

Submitted Manuscript

[M1] Jingfeng Xiong⁺, **Jiacan Li**⁺, Yanchen Wu, Kexin Mu, Hanqing Ma, Yaojin Xu, Zhuoyang Ding, Xinying Li, ChuHeng Chen, and Chenglong Fu*, “Human State Velocity Field Mapping for Exoskeleton Assistive Control.” Manuscript submitted to *Nature Communications*.

⁺ co-first authors; * corresponding author.

Patents and Patent Applications

- [P1] China, publication no. [CN121081241A](#), “Control methods, devices, electronic equipment and storage media.” **Granted. Inventors:** Chenglong Fu, Jingfeng Xiong, **Jiacan Li**, Chuheng Chen, Yanchen Wu, Xinying Li, Youran Xi, and Yaojin Xu.
- [P2] China, publication no. [CN121733493A](#), “An exoskeleton control method, system, electronic equipment and computer program product.” **Granted. Inventors:** Chenglong Fu, Xinying Li, Jingfeng Xiong, Xiaohe Yu, Yanchen Wu, and **Jiacan Li**.
- [P3] China, publication no. [CN120459594A](#), “Resistance exoskeleton mechanism control method, resistance exoskeleton mechanism and bottom controller.” **Published patent application. Inventors:** Chenglong Fu, Shiquan Yu, Yuquan Leng, Guangshan Wang, Xinying Li, **Jiacan Li**, Wenbing Zhuang, and Jingfeng Xiong.

Research Experience

Tactile Friction Estimation for Dexterous Grasping

National University of Singapore | Singapore | Advisor: Xingyu Liu

- Built an NVIDIA Isaac Lab simulation environment for a LEAP Hand equipped with GelSight Mini fingertip sensors.
- Developed a tactile-image-based estimator of local friction coefficients; incorporated the estimates into reinforcement learning (RL) to inform grasp planning and control.

Jun 2026 – Present

Unified Exoskeleton Control for Multi-Terrain Assistance

Southern University of Science and Technology | Shenzhen, China | Advisor: Chenglong Fu

- Developed a unified neural network-based approach to inform exoskeleton assistance from wearer kinematics across level walking, ramps, and stairs, without explicit inverse dynamics modeling.
- Deployed a lightweight model on embedded hardware for real-time inference.
- **Research outputs:** Co-first author of submitted manuscript [M1]; co-inventor of granted patent [P1].

May 2025 – Aug 2026

Gait Estimation for Post-Stroke Rehabilitation

Southern University of Science and Technology | Shenzhen, China | Advisor: Chenglong Fu

- Developed and deployed a neural network-based framework to estimate affected-limb kinematics in real time from contralateral limb motion for lower-limb exoskeleton-assisted stroke rehabilitation.
- **Research output:** Co-inventor of granted patent [P2].

Jan 2026 – Apr 2026

Adaptive Resistance Control for Exercise Exoskeletons

Southern University of Science and Technology | Shenzhen, China | Advisor: Chenglong Fu

- Trained deep learning models to estimate electromyographic (EMG) signals from inertial measurement unit (IMU) data and used these muscle activity estimates to guide adaptive exoskeleton resistance during exercise.
- **Research output:** Co-inventor of published patent application [P3].

Jan 2025 – Mar 2025

Selected Projects

- SplatSLAM: Monocular 3D Reconstruction

Jun 2026

 - Integrated MAST3R-SLAM with 3D Gaussian Splatting for monocular reconstruction and novel view synthesis; implemented camera-pose conversion and COLMAP-compatible data export.
 - Applied statistical outlier removal to reduce rendering floaters, pruning 7.3% of points in a test scene; evaluated the effects of capture coverage and training duration on reconstruction quality and overfitting.
- ROS 2-Based Mobile Manipulator with Multimodal HRI

Jan 2026

 - Built a ROS 2/Gazebo mobile manipulation system integrating OpenCV target detection, MoveIt 2 grasping, and Nav2 navigation for autonomous pick-and-deliver tasks.
 - Implemented gesture and voice interfaces with MediaPipe and Vosk; coordinated missions through state machines and multithreaded ROS 2 execution to maintain responsive interaction during navigation.
- 6-DOF Robotic Arm Automated Drawing System

May 2025

 - Converted images into drawing trajectories using OpenCV skeletonization and waypoint sequencing; derived analytical inverse kinematics from D–H parameters for a 6-DOF manipulator.
 - Designed a spring-loaded compliant pen holder for passive contact regulation; mapped planar paths onto cylindrical surfaces and validated motion in Simscape Multibody before physical execution.
- Agricultural Robot (ROBOCON 2024)

Jul 2024

 - Designed a custom gripper for rice harvesting.
 - Integrated robot hardware, including soldering serial-port connections and custom printed circuit boards (PCBs) for hardware communication.

Grants, Awards & Honors

- Outstanding Student Award

2024, 2026
- Special Funds for the Cultivation of Guangdong College Students’ Scientific and Technology Innovation.

2025

“Climbing Program” Special Funds; CNY 20,000; Leader.
- China Int’l College Students Innovation Competition

2025

National Bronze Award and Provincial Gold Award.
- Second Prize Scholarship (Top 8%, twice)

2024, 2025
- Outstanding Student Leader Award

2025
- ROBOCON China Championship

2024

National First Prize (Operation Challenge); National Second Prize (Theme Contest); two National Third Prizes (Biomimetic Legged Robot).
- Outstanding Volunteer (Top 3%)

2024

Skills & Additional Information

- Programming:** Python, C++, MATLAB
- Robotics & Simulation:** ROS 2, MuJoCo, NVIDIA Isaac Lab, Gazebo
- Machine Learning:** PyTorch
- Computer Vision & Visualization:** OpenCV, RViz2
- Embedded Systems:** STM32, Arduino
- CAD & CAE:** SolidWorks, AutoCAD, Abaqus
- English:** TOEFL iBT: 5.5/6 (110/120 equivalent)
- Certifications:** AHA HeartSaver®, Red Cross Instructor
- Interests:** Go (Board Game), Amateur Level 1

Leadership & Service

- SUSTech Red Cross — Director, Organization Department.** Managed 144.5 volunteer service hours; organized 5+ first aid and emergency response training sessions as an instructor; led 10+ large-scale events and competitions, including carnivals and forums.
- Shenzhen Nanshan District Emergency Rescue Association — Instructor.** Established certification frameworks for first aid instructors and responders.
- Campus First Aid Responder.** Handled 20+ medical emergencies, including epilepsy, fractures, and trauma.
- Additional Leadership:** Student Representative, Dept. of Mechanical & Energy Engineering; Head of Operations, Robotics Club.